

AI Services and Solutions by Domain, Paradigm & Vertical

Domains / Industries

- **Healthcare & Life Sciences** – AI-based diagnostics, prescription analysis, symptom triage, second-opinion systems, medical image analysis, mental-health support, AR/VR training, appointment booking and patient portals.
- **Banking & Financial Services** – Document OCR/ICR, virtual financial assistants with voice biometrics and fraud detection, multilingual chatbots for basic account services.
- **E-commerce & Retail** – Conversational commerce engines, intelligent negotiation, recommendation systems, AI-first retail OS with voice assistants, multi-branch inventory and credit management.
- **Real Estate & Property Management** – Decision-support platform combining energy optimisation, predictive maintenance, tenant satisfaction prediction and rent optimisation.
- **Education & Training** – 3D anatomy exploration with AR/VR, surgical simulations with haptics, AI mentors and evaluators to support medical students.
- **Cross-Vertical Platforms** – Multimodal, multilingual assistants (LUNA, 1-Tap App) and MLOps pipelines supporting multiple business domains.

Technology Paradigms

- **Natural-Language Processing (NLP)** – Differential diagnosis, chatbots for patients, students and banking customers, multi-turn conversational agents, sentiment analysis, question-answering engines.
- **Computer Vision** – Medical imaging (CT/MRI/X-ray) analysis, pathology heatmap localisation, AR/VR 3D models, product image search.
- **Optical Character Recognition (OCR/ICR)** – Handwritten prescription reader, banking document extraction (Bangla + English), digital form processing.
- **Speech & Voice Technologies** – Automatic speech recognition (ASR), text-to-speech (TTS), voice biometrics, real-time translation for 40+ languages, call-centre whisper agents.
- **Recommendation & Decision Systems** – Treatment and drug recommendations, product suggestions, capital-stuck estimation, maintenance prioritisation, energy and rent optimisation.
- **Predictive Analytics & Machine Learning** – Disease prediction, energy consumption forecasting, equipment failure prediction, tenant satisfaction forecasting, user churn prediction.
- **Human-in-the-Loop (HITL) Systems** – Confidence-based routing of OCR/ICR results and AI diagnoses to human reviewers, maker-checker workflows.
- **MLOps & AI Engineering** – Custom pipelines for data ingestion, model training and deployment (Kubernetes, MLflow), vector databases (Pinecone/Qdrant), CI/CD, monitoring.
- **Augmented & Virtual Reality** – Immersive anatomy labs, haptic-enabled surgical simulations with AI scoring.

Business Verticals / Use-Case Categories

- **Patient Engagement & Clinical Decision Support** – Symptom checkers, cancer pre-screening, second opinions, medication guidance, appointment booking, doctor recommendation.
- **Document Processing & Workflow Automation** – Prescription OCR, bank form extraction, report upload/download, lab result analysis.
- **Customer Service & Voice Assistants** – Multi-domain chatbots, call-centre whisper agents, order tracking, scheduling, voice biometrics for authentication.

- **Training & Education** – 3D anatomy modules, AI mentors, surgical simulation, interactive case studies.
- **Business Analytics & Decision Support** – Real-estate energy optimisation, predictive maintenance, rent optimisation, retail business health scoring and capital estimation.
- **Cross-Vertical AI Platforms** – LUNA and 1-Tap App frameworks enabling multi-modal, multi-channel AI experiences across medical, banking, retail and personal domains.

Consolidated Service Table

Service Name	Key Technologies	Maturity	Target Users	Quantified Benefits
Healthcare Diagnostic Assistant	CNNs, GPT-4V, BERT-style clinical NLP, OCR transformer, FHIR integration	Live / Advanced MVP	Radiologists, doctors, patients	85% reduction in screening time; 40% increase in early detection; 95.6% OCR accuracy
Multimodal Conversational Platform (LUNA)	Claude/GPT-4, Whisper ASR, Google TTS, LangChain, voice biometrics	Live	E-commerce customers, bank customers, patients	22% conversion lift; 70% agent deflection; 3.2× user growth
Bank Document OCR/ICR Engine	LLaMA-vision model, Tesseract, confidence scoring, maker-checker workflow	In Development / Pilot	Bank back-office staff, compliance teams	70–80% faster processing; 95%+ accuracy (EN)
AI-First Retail OS Copilot	Next.js, TanStack Query, Web Speech API, z-ai-web-sdk	Concept / Early MVP	SME shop owners	Business health scoring, offline-capable operations, credit control
Real-Estate Decision Support Platform	Azure Synapse, XGBoost, Isolation Forest, hedonic pricing, React	Proposal / Proof of Concept	Property managers, sustainability teams	10–15% projected reduction in operating costs; 8–12% energy savings

Service Name	Key Technologies	Maturity	Target Users	Quantified Benefits
AR/VR Medical Learning Suite	Unity, C#, haptic controllers, eye-tracking vision models, NLP	Live / Pilot	Medical students, residents, faculty	78% knowledge retention improvement; 50% fewer complications
Diagnostic Centre Digital Transformation & Chatbot	Next.js, Express, PostgreSQL, Claude/Groq APIs, Twilio, AWS S3	Live (Phase-1) / MVP (Phase-2)	Patients, clinic staff, administrators	60% reduction in scheduling workload; >50% portal adoption
1-Tap App AI Chatbot	GPT-based models, multi-language NLP, backend API integration	Live / MVP	Consumers using 1-Tap service platform	≥85% chatbot accuracy; <3 s response time; >90% user satisfaction
Power BI & Automate Analytics (Jotax)	Power BI, DAX, Power Automate, custom Finnish week logic	Live	Finance analysts	80% reduction in reporting time; elimination of €42K quarterly variance

Note: Descriptions for each service are provided in the case-study sections rather than in this table to adhere to the guideline of avoiding long sentences inside tables.

Detailed Case Studies

Each case study uses anonymised project names and clients due to confidentiality agreements. A high-level architecture diagram accompanies the narrative description. Status/Remarks indicate current delivery stage (Live, MVP, Pilot, etc.).

Project A – Healthcare Diagnostic Assistant

Client / Stakeholder: Large regional hospital network and telemedicine provider (anonymous due to NDA).

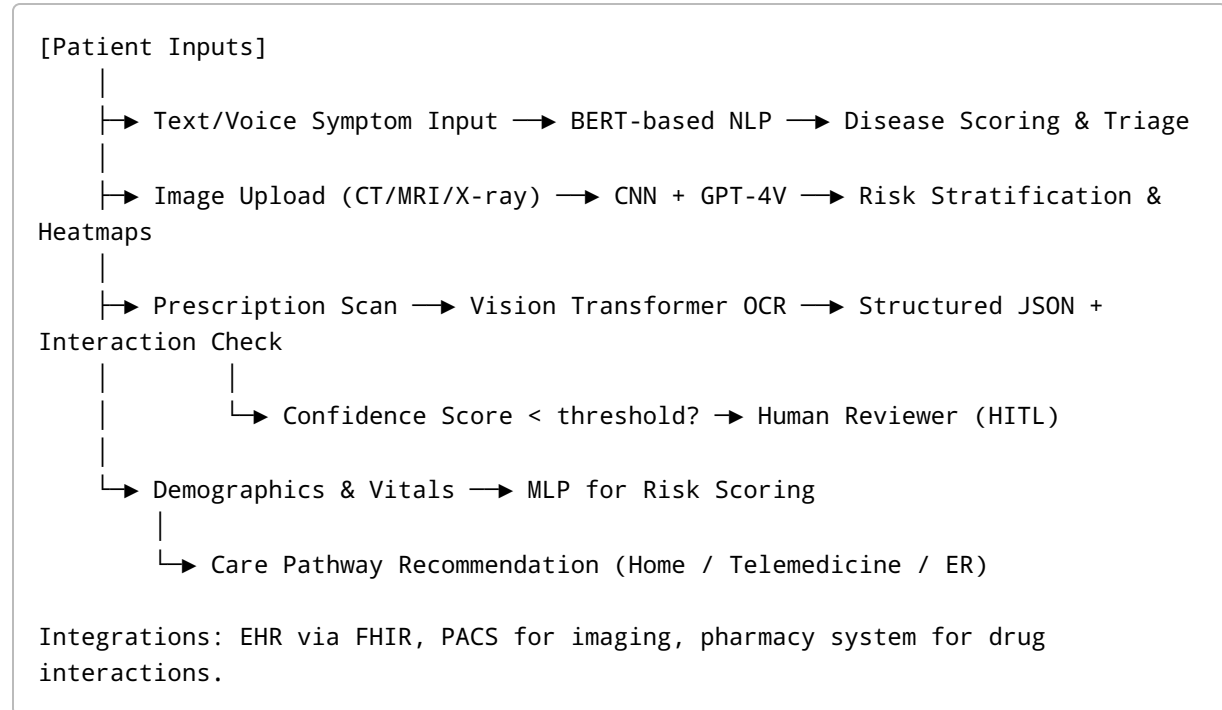
Status / Remarks: Live for prescription OCR, symptom analysis and differential diagnosis; advanced MVP for multi-modal imaging and cancer pre-screening modules.

Pain Points:

- Radiologist workload increase of 30–50% and interpretation errors of 3–5%.
- Rural areas face a **65% shortage** of oncologists causing 4-6 week diagnosis delays.
- Illegible handwritten prescriptions lead to medication errors.

- Primary care doctors spend 40% of their time triaging cases that may not require expert intervention.

Solution Architecture:



Technical Differentiators:

- **Multimodal processing** (text, voice, imaging, scanned documents) unified under a single platform.
- **Custom Vision Transformer** for Bangla/English prescription OCR achieving **95.6% accuracy**.
- **Confidence-based HITL**: results below confidence threshold are routed to human reviewers.
- Integration with **PACS/EHR** systems using HL7 FHIR APIs.

Business Impact:

- **Screening time reduced** from 3 weeks to 24 hours (85% reduction).
- **Radiologist workload decreased** by 35% via pre-filtering negative cases.
- **40% increase** in early-stage cancer detection.
- **Prescription error reduction** through OCR accuracy of 95.6% and drug-interaction checks.

Team Size & Duration:

~8 engineers (AI/ML, frontend, backend), 2 data scientists, 2 QA testers; 9-month development for initial modules, ongoing iterative enhancements.

Project B – Multi-Channel Conversational Platform

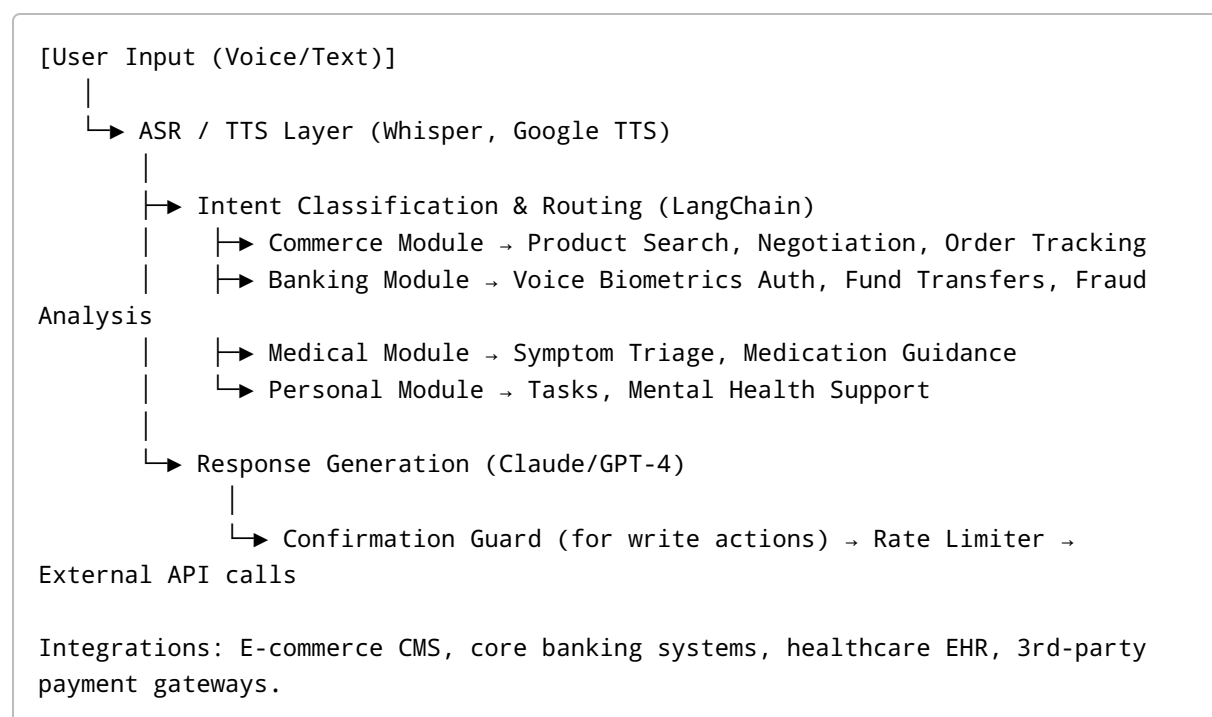
Client / Stakeholder: Consumer goods marketplace, regional bank, and healthcare provider (multi-tenant platform) – anonymised.

Status / Remarks: Live across e-commerce and medical verticals; pilot in banking vertical.

Pain Points:

- Call-centre cost escalation of **\$1.30–1.80 per minute** with average handle time increasing 15% year-over-year.
- 72% of consumers prefer support in their native language, causing a **40% loss of international leads** due to language barriers.
- Cart abandonment of 70% in e-commerce because of lack of real-time assistance.
- Banking call volume dominated by low-value tasks (password resets, balance inquiries) amounting to 60% of calls.

Solution Architecture:



Technical Differentiators:

- **Voice biometrics** with **99.1% authentication accuracy** and 3-second verification.
- **Multilingual support** for Bangla, English and Banglish with real-time translation across **40 languages**.
- **Reinforcement-learning negotiation engine** for dynamic discounting and cart recovery.
- **Sentiment-based escalation prediction** that anticipates call escalation 90 seconds ahead with **72% accuracy**.

Business Impact:

- **22% conversion lift** and **+12% average order value** in voice-assisted commerce sessions.
- **70% deflection** of support interactions away from human agents.

- Banking: **60% reduction** in call volume (approx. **\$12M annual savings**) and **45% decrease** in social-engineering fraud.
- Medical: **4.8/5** patient satisfaction for guidance accuracy.

Team Size & Duration:

12 engineers (AI, voice, frontend, backend), 3 data scientists, 2 UX designers; 6-month initial build with continuous enhancements.

Project C – Document OCR/ICR for Banking

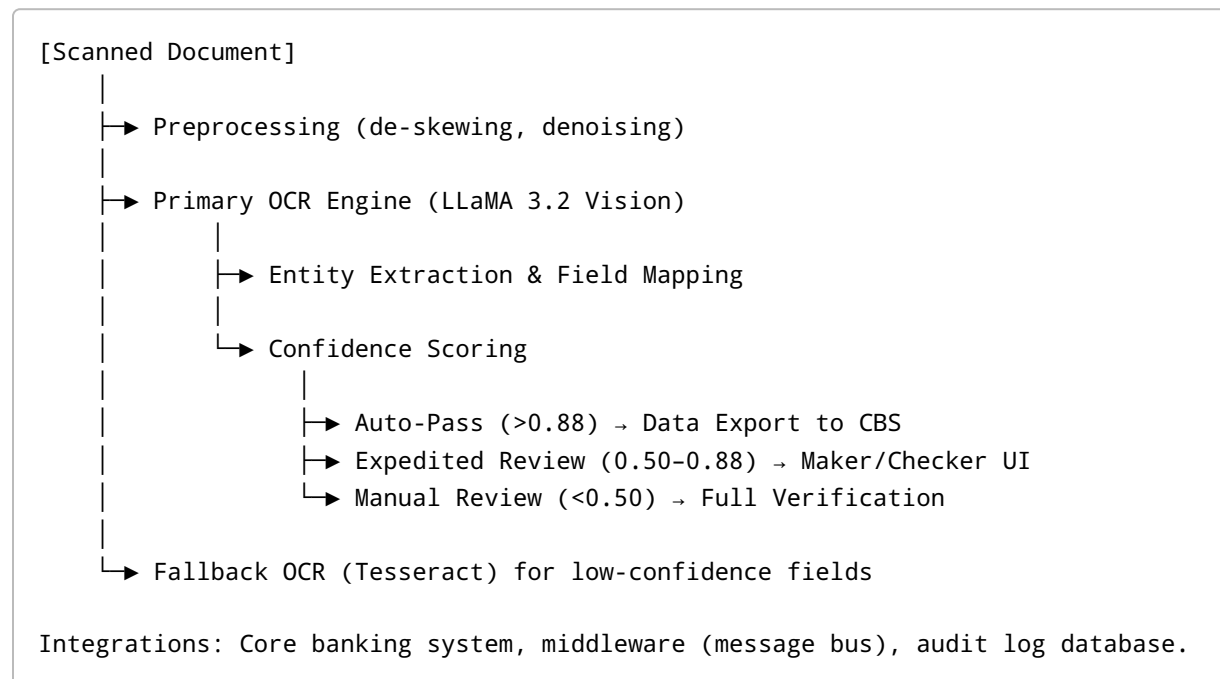
Client / Stakeholder: National commercial bank (anonymous).

Status / Remarks: Pilot implementation; POC validated and being integrated with core banking system.

Pain Points:

- Manual data entry from account opening forms requires **15–20 minutes per document** with error rates of **2–5%**.
- Scans contain mixed Bangla-English text, handwriting, strikethroughs and inconsistent quality.
- Lack of audit trail and maker-checker workflow required for regulatory compliance.

Solution Architecture:



Technical Differentiators:

- **Hybrid OCR pipeline** combining state-of-the-art vision models with traditional OCR for fallback.
- **Confidence-based routing** enabling HITL review and maker-checker approval flows.

- Containerised deployment for on-prem hosting (Docker/Kubernetes) to meet data-sovereignty requirements.
- Immutable audit log ensuring compliance with banking regulations.

Business Impact:

- **95%+ English OCR accuracy, 85%+ Bangla accuracy** (pilot results).
- **70–80% faster** document processing—average review time <3 minutes per form.
- Automated audit trail satisfying regulatory requirements.

Team Size & Duration:

6 AI engineers, 2 backend developers, 1 compliance officer; 4-month POC followed by ongoing integration.

Project D – AI-First Retail Operating System

Client / Stakeholder: SME software provider for retail merchants (anonymous).

Status / Remarks: Concept and early MVP; modules built for demonstration and architecture design.

Pain Points:

- Most ERPs treat AI as an add-on; SME shop owners need a **Bangla-first** system that works offline and handles voice commands.
- Friction in data entry, lack of integrated analytics, poor business visibility (profit, credit, stock).
- High connectivity costs and limited tech literacy among small retailers.

Solution Architecture:

```
[Next.js Frontend]
|
|→ Persistent AI Drawer & Voice AI (Web Speech API)
|   |→ z-ai-web-dev-sdk → LLM interface (Claude/Grok)
|
|→ Data Layer
|   |→ @/services - Axios REST calls (Two-layer rule)
|   |→ @/hooks/api - TanStack hooks + offline queue
|
|→ Business Modules
|   |→ Analytics & Health Score → Business KPIs, capital-stuck
estimation
|   |→ Inventory → Multi-tier pricing, dead stock detector
|   |→ Credit Control → Risk scoring, collection priority
|   |→ Sales/POS → AI margin meter, multi-account payments
|   |→ System Core → Offline sync, audit logs, role-based access
|
|→ Multi-tenancy, i18n (EN/BN) via next-intl
```

Back-end: Separate REST API server; AI safety via confirmation guard and rate limiter.

Technical Differentiators:

- **Bangla-first user experience** with dual-language keys (English/Bangla) and offline capability.
- **Persistent AI Copilot** integrated into the OS (not just a chatbot) supporting voice commands and quick actions.
- **Two-layer data rule** ensuring separation between service calls and hooks for maintainability.
- AI safety components (confirmation guard, rate limiter) to prevent unintended actions.

Business Impact (Projected):

- Improved user adoption due to **native language** support and frictionless interactions.
- **Dead stock detection** and capital-stuck estimation help retailers free up working capital.
- Business Health Score provides visibility, enabling smarter credit and inventory decisions.

Team Size & Duration:

8 full-stack engineers, 1 UX designer, 2 AI engineers; early research and architecture design spanning 3 months.

Project E – Real-Estate Decision Support Platform

Client / Stakeholder: Large property-management firm (anonymous).

Status / Remarks: Proposal and proof-of-concept demonstration.

Pain Points:

- Data siloed across **7+ systems** (Greenview, Fast2, Vitec, MS CRM, IoT sensors).
- Reactive decision-making in energy management, maintenance and tenant satisfaction leading to increased costs and tenant churn.
- No unified analytics or predictive insights for optimizing rents and maintenance schedules.

Solution Architecture:

```
[Data Ingestion]
├─> Azure Synapse / Databricks ETL → unified warehouse
│   └─> Data from CRM, ERP, IoT, tenant surveys
├─> ML Modeling
│   └─> Energy Baseline (XGBoost) → MAPE <5%
│   └─> Anomaly Detection (Isolation Forest)
│   └─> Predictive Maintenance (Survival Analysis + Random Forest)
│   └─> Tenant Satisfaction & Rent Optimisation (Hedonic Pricing)
```

- ↳ Action Engine
 - ↳ Prioritised Recommendations (impact scores)
 - ↳ Role-Based Dashboards (React)

Authentication: OAuth 2.0; Integration with existing property-management systems.

Technical Differentiators:

- **Unified data lake** that aggregates disparate sources and supports advanced analytics.
- **Multi-model ML pipeline** combining forecasting, anomaly detection and survival analysis.
- **Hedonic pricing models** for rent optimisation and vacancy risk prediction.
- Modular design enabling integration with future IoT sensors and property management platforms.

Business Impact (Projected):

- **10–15% reduction** in operational and energy costs; **8–12% energy savings**.
- **5% improvement** in tenant satisfaction scores due to proactive maintenance and fair rent settings.
- Better allocation of maintenance budgets and reduced vacancy periods.

Team Size & Duration:

4 data scientists, 2 software engineers, 1 data engineer; 2-month POC with potential for scale-up.

Project F – AR/VR Medical Learning Suite

Client / Stakeholder: Medical university and teaching hospital consortium (anonymous).

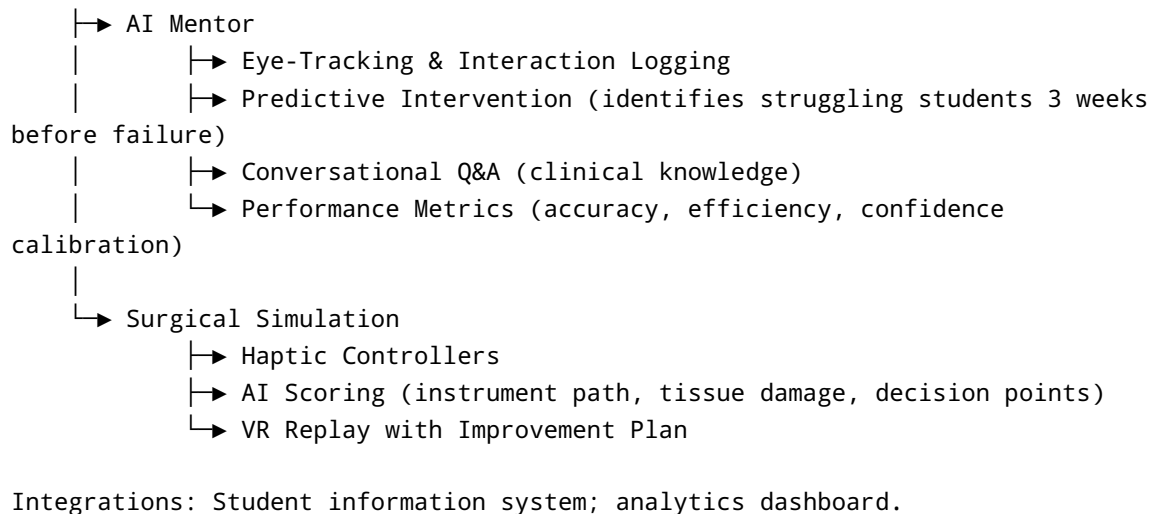
Status / Remarks: Deployed in pilot programmes; expanding to more institutions.

Pain Points:

- Cadaver shortage (1 cadaver per 6–8 students; ideal 1:4) and high lab costs (~\$2.4 M).
- Traditional lectures lead to **60% knowledge loss** within a year; 40% of students struggle with 3D spatial reasoning.
- Instructor evaluation bias causing **35% variance** in student assessments.

Solution Architecture:

- [AR/VR Hardware (Meta Quest, HoloLens)]
 - ↳ Anatomical 3D Models (Unity Engine)
 - ↳ Layered Dissection
 - ↳ Dynamic Physiology Simulation
 - ↳ Pathology Generator (500+ disease states)
 - ↳ Multiplayer Collaboration (10 users)



Technical Differentiators:

- **Immersive 3D and haptic simulation** enabling hands-on practice without cadavers.
- **AI Mentor** uses eye-tracking and interaction logs to predict and intervene before students fail (89% accuracy).
- **Standardised evaluation** eliminates instructor bias and provides consistent scoring.
- Multiplayer collaboration fosters peer learning and remote participation.

Business Impact:

- **78% improvement** in knowledge retention after 3 months.
- **60% reduction** in per-student training cost over three years.
- **50% fewer complications** in residents' first 10 procedures.
- **70% reduction** in faculty Q&A workload.

Team Size & Duration:

7 Unity/AR developers, 2 AI specialists, 2 UX researchers; 12-month development with iterative module releases.

Project G – Diagnostic Centre Digital Transformation

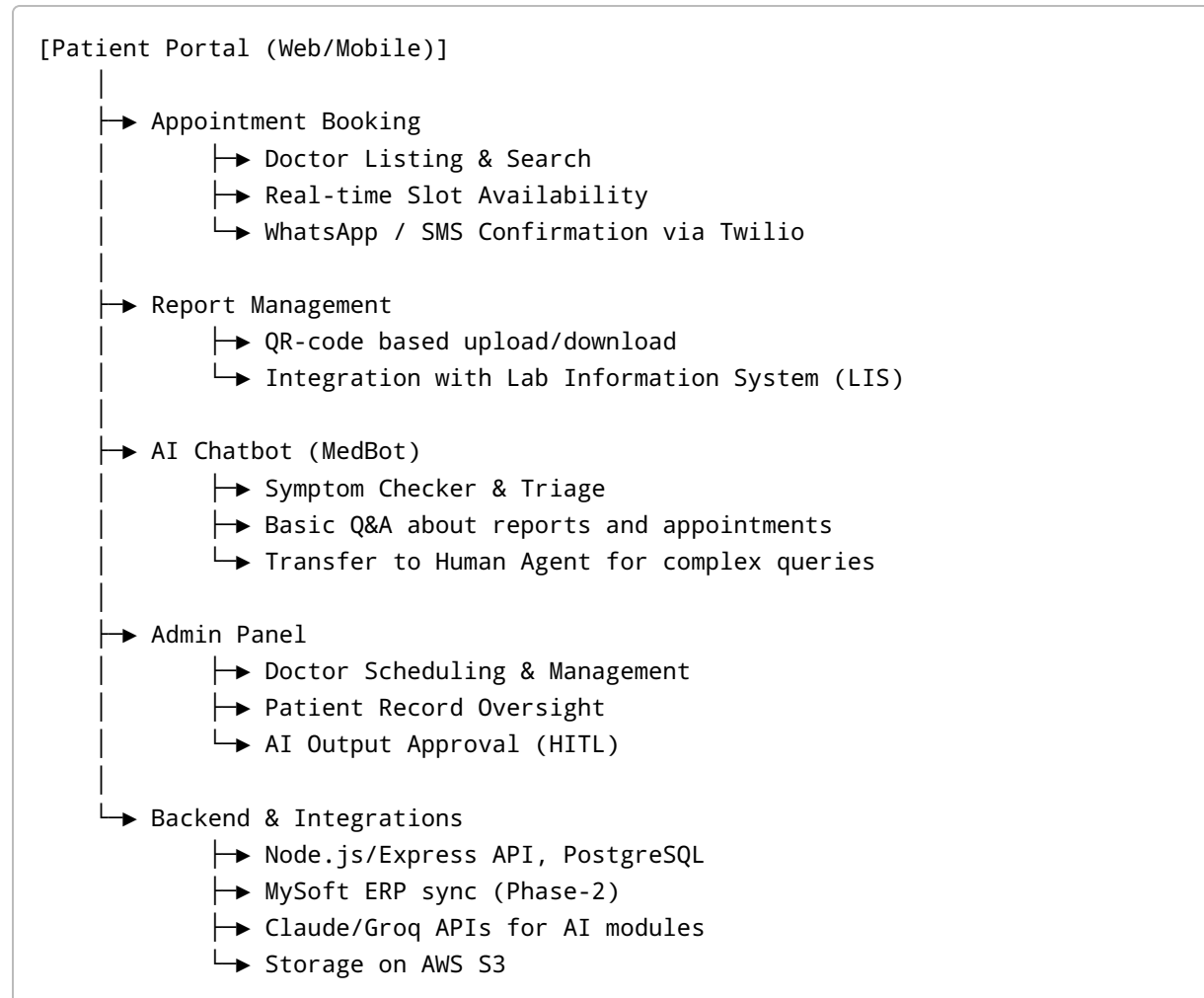
Client / Stakeholder: Urban diagnostic & consultation centre (anonymous).

Status / Remarks: Phase-1 (8-week MVP) delivered and live; Phase-2 features under development.

Pain Points:

- Manual appointment booking and scheduling; no 24/7 self-service for patients.
- Paper-based reports; patients cannot access medical records digitally.
- Lack of integration with the existing MySoft ERP; data duplication and inefficiencies.
- No AI-powered triage or symptom checker for initial guidance.

Solution Architecture:



Technical Differentiators:

- **Quick-reply chips and typing indicators** provide a conversational user experience with low cognitive load.
- **OTP-protected patient portal** ensures secure access to medical records.
- **AI-powered triage** leverages Claude/Groq models for symptom analysis and doctor recommendations.
- **Integration roadmap** with existing ERP and EMR systems (planned in Phase-2).

Business Impact:

- **60% reduction** in staff scheduling workload; bookings now under 3 minutes.
- **>50% portal adoption** after launch; paper-less report handling reduces administrative overhead.
- Patient satisfaction improved through 24/7 access and real-time notifications.

Team Size & Duration:

5 full-stack developers, 2 AI engineers, 1 project manager; delivered MVP in 8 weeks with ongoing Phase-2 development.

Comparative Analysis & Synergies

Recurring Architectural Patterns

1. **Confidence-based HITL Workflows** – Both the diagnostic assistant and banking OCR systems route low-confidence outputs to human reviewers, ensuring safety and regulatory compliance.
2. **Multimodal Design** – Solutions integrate various inputs (text, voice, image, scanned documents) and outputs (voice, text, heatmaps).
3. **Multilingual & Localisation Support** – Many solutions cater to Bangla, English and Banglish speakers; retail OS and LUNA integrate dual-language keys and real-time translation.
4. **Modular Microservices** – RESTful APIs, two-layer data access patterns (services vs. hooks), and integration via standardized protocols (HL7 FHIR, OAuth 2.0).
5. **Offline & Low-Connectivity Resilience** – Retail OS uses offline sync queues; AI chatbots handle network outages gracefully.

Unique Technical Risks & Mitigation Strategies

Risk	Affected Projects	Mitigation Strategy
Bangla Handwriting OCR	Diagnostic assistant, banking OCR	Custom vision transformer trained on mixed Bangla/English datasets; confidence thresholds trigger manual review; fallback to Tesseract for edge cases.
On-Prem Deployment Constraints	Banking OCR	Containerised deployment (Docker/Kubernetes); local compute clusters; secure API integration with bank's core system.
Data Privacy & Compliance	Healthcare projects, bank OCR	HIPAA/GDPR compliance: anonymisation, encryption, audit logs, role-based access control; internal review processes.
Multilingual Processing & Translation	LUNA, retail OS, 1-Tap Chatbot	Real-time translation services; dual-language data models; fallback to human agents for low-confidence translations.
Offline Operation & Low Connectivity	Retail OS, diagnostic centre	Offline queues and background sync; local caching; minimal dependencies on external services for critical functions.
Domain-specific Logic	Power BI analytics (Finnish calendar), real-estate models	Custom DAX functions implementing Finnish week numbering; hedonic pricing models trained on local market data.

Risk	Affected Projects	Mitigation Strategy
MLOps Complexity	Cross-vertical AI services	Automated versioning, CI/CD for models, GPU-optimised inference; monitoring and rollback mechanisms.

Maturity / Readiness Matrix

Service	Concept	MVP	Pilot	Live	Remarks
Healthcare Diagnostic Assistant		✓	✓	✓	Imaging & OCR modules advanced; ongoing enhancements
Multimodal Conversational Platform (LUNA)		✓	✓	✓	Live across commerce and healthcare; banking pilot
Banking Document OCR/ICR			✓		POC completed; integration underway
AI-First Retail OS Copilot	✓	✓			Modules under early MVP; architecture defined
Real-Estate Decision Support	✓		✓		Proof-of-concept; awaiting approval
AR/VR Medical Learning Suite			✓	✓	Pilots launched in multiple institutions
Diagnostic Centre Digital Transformation		✓		✓	Phase-1 live; Phase-2 under development
1-Tap App AI Chatbot		✓		✓	Live with ongoing fine-tuning
Power BI & Automate Analytics		✓		✓	Delivered and operational

Potential Synergies

- OCR Engine Reuse:** The **prescription OCR** developed for healthcare can be adapted to process **bank forms**, leveraging the same Bangla/English mixed-script recognition and confidence scoring.
- LUNA Framework Extension:** LUNA's modular architecture and translation capabilities could be customised for **real-estate tenant portals**, providing voice-assisted maintenance requests and rent queries.
- Unified AI Orchestration:** A central MLOps pipeline can manage models across healthcare, banking, retail and real-estate, enabling consistent deployment, monitoring and resource sharing.
- Cross-domain Data Insights:** Combining energy consumption and predictive maintenance data with financial analytics could lead to **comprehensive sustainability dashboards** for corporate clients.
- Shared Voice Biometrics:** The voice-authentication module from banking can secure patient portals and retail apps, ensuring consistent user verification across verticals.

6. **Multi-modal Medical Models:** The image-analysis engine (CNN + GPT-4V) could be repurposed for **industrial inspection** or **agricultural disease detection**, opening new markets.
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Lessons Learned & Expansion Opportunities

1. **Human Oversight is Essential:** HITL workflows ensure safety and build trust in AI systems. Projects that incorporate confidence thresholds and manual review achieve higher adoption and regulatory acceptance.
2. **Local Language Support Drives Adoption:** Bangla-first interfaces and multilingual chatbots significantly improve user engagement in the Bangladeshi context. Future expansions should invest in other regional languages to capture broader markets.
3. **Modularity Enables Reusability:** Designing AI solutions as independent modules (OCR, ASR, NLP, recommendation) allows them to be recombined for new verticals, accelerating time-to-market for future products.
4. **MLOps & Governance:** Early investment in MLOps pipelines simplifies deployment and monitoring of models across multiple domains. A central governance framework ensures ethical AI use and compliance with data-protection regulations.
5. **User Experience Matters:** Conversational UI elements like quick-reply chips, typing indicators and voice feedback make AI systems accessible to non-technical users and reduce friction.
6. **Data Quality & Domain Expertise:** Success depends on high-quality annotated data and close collaboration with domain experts (doctors, bankers, property managers). Incorporating domain knowledge in model design greatly enhances accuracy and usability.

Potential Expansion:

Future roadmaps may include integration with wearable sensors for remote patient monitoring, expansion of LUNA to new languages (Hindi, Arabic), adoption of generative agents in supply-chain management, and cross-industry analytics platforms. By leveraging existing modular components (e.g., OCR engine, voice biometrics), the portfolio can scale efficiently into new sectors and geographies.

Appendix

Glossary

Term	Definition
ASR	Automatic Speech Recognition – converts spoken words into text.
BERT	Bidirectional Encoder Representations from Transformers – an NLP model for understanding language context.
CNN	Convolutional Neural Network – neural network architecture commonly used for image analysis.
EHR / EMR	Electronic Health Record / Electronic Medical Record – digital system for storing patients' medical histories.

Term	Definition
FHIR	Fast Healthcare Interoperability Resources – standard for exchanging healthcare information electronically.
HITL	Human-in-the-Loop – workflow where humans validate or correct AI outputs.
LIS	Laboratory Information System – software managing lab processes and test results.
LLM	Large Language Model – AI model trained on vast text corpora to generate or understand language.
NLP	Natural-Language Processing – field of AI focused on interactions between computers and human language.
OCR/ICR	Optical Character Recognition / Intelligent Character Recognition – technology that converts images of text into machine-encoded text.
PACS	Picture Archiving and Communication System – medical imaging storage and retrieval system.
POC	Proof of Concept – initial demonstration to validate feasibility.
STT/TTS	Speech-to-Text / Text-to-Speech – technologies converting speech into text and vice versa.
XGBoost	Gradient boosting algorithm widely used for tabular data forecasting.

Technology Stack References

- **AI & ML Frameworks:** Anthropic Claude, Groq (Llama-3), GPT-4V, XGBoost, Isolation Forest, survival analysis models, Unity engine, LangChain.
- **Frontend:** React, Next.js, Tailwind CSS, TanStack Query, Zustand.
- **Backend:** Node.js (Express), NestJS, FastAPI, Python.
- **Data & Storage:** PostgreSQL, MongoDB, Redis, AWS S3, MinIO, Pinecone/Qdrant, Azure Synapse, Databricks.
- **Messaging & Integration:** RabbitMQ, REST APIs, HL7 FHIR, OAuth 2.0, Twilio, Web Speech API.
- **DevOps/MLOps:** Docker, Kubernetes, MLflow, CI/CD pipelines, GitHub Actions, Power BI, Power Automate.